Q1: Top Discounted Products

Business Problem

Identify products offering the highest discounts to attract customers and boost sales.

Insight

- Products like **Dukes Waffy Wafers** and **Chef's Basket Pasta** have **~50–51% discounts**, which are extremely high.
- Multiple variants of the same brand (chocolate, strawberry, orange) are heavily discounted.
- This indicates **bulk discounting strategy on similar product lines**.

Business Impact

- High discounts can significantly **increase customer purchases**.
- However, it may **reduce profit margins** if not controlled.
- Repeated discounting on same brand suggests possible **overstock or low demand**.

Recommendation

- Use high discounts strategically for **slow-moving inventory**.
- Avoid heavy discounts on already popular products.
- Introduce **bundle offers** instead of direct price cuts to maintain margins.

Q2: High MRP but Out-of-Stock Products

Business Problem

Identify expensive products that are unavailable, leading to potential revenue loss.

Insight

- Products like:
  - **Patanjali Cow's Ghee (₹565)**
  - **MamyPoko Diapers**
  - **Aashirvaad Atta**
    are **out of stock despite high demand categories**.
- Same product appears in **multiple categories**, showing strong demand across segments.

Business Impact

- Out-of-stock high-value items lead to **direct revenue loss**.
- Customers may switch to competitors → **loss of customer trust**.
- Indicates **poor inventory management for essential goods**.

Recommendation

- Maintain **minimum stock levels** for essential and high-MRP products.
- Use demand forecasting to **prevent stockouts**.
- Prioritize restocking of **high-revenue-generating items**.

🔹 Q3: Revenue by Category

Business Problem

Analyze which product categories generate the most revenue to optimize business strategy.

Insight

- **Top performing categories:**
  - Cooking Essentials & Munchies (~₹337K)
  - Personal Care & Paan Corner (~₹270K)
  - Packaged Food / Chocolates (~₹224K)
- **Low performing categories:**
  - Fruits & Vegetables (~₹10K)
  - Meats (~₹20K)
- Some categories have **identical revenues**, indicating:
  - Possible **data duplication**
  - Or shared product entries across categories

Business Impact

- Business is heavily dependent on **packaged and essential goods**.
- Fresh categories (fruits, meats) are **underperforming**.
- Risk of **over-dependence on few categories**.

Recommendation

- Invest more in **high-performing categories** (ads, offers, visibility).
- Improve supply chain for **low-performing fresh categories**.
- Investigate duplicate revenue patterns for **data accuracy**.

Q4: High MRP Products with Low Discount

Business Problem

Identify expensive products that have low or no discounts to understand pricing strategy and its impact on sales.

Insight

- Many **high-value products (₹500–₹1250)** such as:
    - Cooking oils (Dhara, Fortune, Saffola)
    - Personal care (L'Oreal, Dove, Sunsilk)
    - Baby & health products (Nestle, Pediasure)
      have **very low or 0% discount**.
- Even essential items like:
    - **Ghee, oil, shampoo, detergent**
      are priced high with minimal discounts.
- Only a few products offer **small discounts (5–9%)**, which is not attractive compared to Q1 (50%+ discounts).

Business Impact

- High prices + low discounts may:
    - Reduce **customer purchase intention**
    - Push customers toward **cheaper alternatives or competitors**
- However, this also indicates:
    - These products may have **strong brand demand**
    - Business is trying to **protect profit margins**

Recommendation

- Provide **moderate discounts (10–20%)** on high-MRP products to increase conversions.
- Use **targeted discounts** instead of flat discounts (festivals, bundles).
- Keep low discounts only for **high-demand branded products**.

Q5: Categories with Highest Average Discount

Business Problem

Identify which product categories offer the highest average discounts to understand promotional strategies.

Insight

- Categories with highest average discounts:
    - **Fruits & Vegetables (~15.46%)**
    - **Meats (~11%)**
    - **Packaged food / Chocolates (~8.3%)**

- Fresh categories (fruits, meat) have **higher discounts** compared to packaged goods.

Business Impact

- Higher discounts in fresh categories indicate:
  - Need to **sell quickly (perishable items)**
  - Avoid **inventory wastage**
- Lower discounts in packaged categories suggest:
  - **Stable demand**
  - Less urgency to push sales

Recommendation

- Continue discounts on **perishable items** to reduce waste.
- Introduce **dynamic pricing** based on shelf life.
- Increase discounts slightly in **packaged food** to boost volume sales.

Q6: Price per Gram (Best Value Products)

Business Problem

Identify which products offer the best value for money by calculating price per gram.

Insight

- Products with **lower price per gram** provide better value to customers.
- Bulk or larger-sized products generally have **lower price per gram**, making them more economical.
- Smaller or premium products tend to have **higher price per gram**, indicating premium pricing.

Business Impact

- Customers are more likely to choose **better value products**, especially in price-sensitive markets.
- Helps identify:
  - **Overpriced products**
  - **Competitive pricing opportunities**

Recommendation

- Promote **low price-per-gram products** as "Best Value Deals".

- Bundle smaller products to **reduce perceived cost**.
- Adjust pricing for high price-per-gram items to stay competitive.

Q7: Product Segmentation by Weight

Business Problem

Categorize products based on weight to understand inventory distribution and customer buying patterns.

Insight

- Products are grouped into:
  - **Low (≤1000g)** → small, daily-use items
  - **Medium (1000–5000g)** → family-size packs
  - **Bulk (>5000g)** → wholesale or heavy usage
- Most products typically fall into **Low and Medium categories**, indicating focus on daily consumption.

Business Impact

- Helps in:
  - Understanding **customer purchase behavior**
  - Designing **packaging strategies**
- Bulk products may:
  - Move slower
  - But generate **higher revenue per sale**

Recommendation

- Promote **medium-size packs** for better balance of price and volume.
- Offer discounts on **bulk purchases** to increase sales.
- Optimize inventory by stocking more **high-demand size categories**.

Q8: Total Inventory Weight per Category

Business Problem

Analyze total inventory weight across categories to understand stock distribution and storage needs.

Insight

- Categories with higher total weight indicate:
  - **Large stock volume**

    - Possibly **high demand or overstocking**
- Categories with lower weight may indicate:
    - **Low inventory levels**
    - Risk of stockouts

Business Impact

- Helps in:
    - Warehouse management
    - Supply chain optimization
- Overstocking increases:
    - Storage cost
    - Risk of wastage (especially perishables)

Recommendation

- Maintain optimal inventory levels using **demand forecasting**.
- Reduce stock in **low-demand heavy categories**.
- Increase stock for **high-demand categories to avoid shortages**.

Q9: Top 3 Products in Each Category by Discount

Business Problem

Identify the top 3 products in each category offering the highest discounts to understand promotional strategies and customer attraction.

Insight

- Many categories have **consistent high discounts (~50%)**, especially:
    - **Biscuits, Cooking Essentials, Packaged Food, Munchies**
- Same products (like *Chef's Basket Pasta* and *Ceres Foods Masala*) appear across multiple categories, indicating:
    - **Cross-category tagging**
    - Or **data duplication**
- Some categories like:
    - **Fruits & Vegetables**
    - **Home & Cleaning**
      have **much lower discounts (16–23%)**

- Personal care categories show **moderate discounts (25–45%)**, not as aggressive as packaged foods.

💼 Business Impact

- High discounts in certain categories indicate:
  - **Aggressive marketing strategy**
  - Focus on **driving volume sales**
- Lower discounts in essential categories suggest:
  - Stable demand
  - Less need for promotions
- Duplicate product presence across categories may:
  - Distort analysis
  - Lead to incorrect business decisions

🚀 Recommendation

- Standardize discount strategies across categories for consistency.
- Avoid excessive discounting on the same products across multiple categories.
- Investigate and fix **duplicate or misclassified product entries**.
- Increase discounts slightly in low-discount categories to improve competitiveness.

Q10: Categories with Above Average Revenue

🧠 Business Problem

Identify categories generating revenue above the average to understand top-performing business segments.

📊 Insight

- High-performing categories include:
  - **Munchies (~₹337K)**
  - **Cooking Essentials (~₹337K)**
  - **Personal Care (~₹270K)**
  - **Paan Corner (~₹270K)**
  - **Chocolates & Candies (~₹224K)**

    - **Packaged Food (~₹224K)**
    - **Ice Cream & Desserts (~₹224K)**
- **Munchies and Cooking Essentials** are the **top revenue drivers**.
- Multiple categories have **exact same revenue values**, indicating:
    - Possible **duplicate entries**
    - Or **same products mapped to multiple categories**

Business Impact

- Business is heavily dependent on:
    - **Packaged and daily-use categories**
- High-performing categories contribute major share of revenue → **core business drivers**
- Duplicate revenue patterns may:
    - Mislead decision-making
    - Overestimate category performance

Recommendation

- Focus marketing and inventory on **top categories (Munchies, Cooking Essentials)**
- Improve data quality by:
    - Removing duplicates
    - Validating category assignments
- Diversify growth by:
    - Improving lower-performing categories
    - Reducing dependency on few categories